DocMind: technical overview

DocMind lets you ask questions about a PDF and get answers with page citations.
It is built with a NestJS API and a React front end.

How uploads work
When you upload a PDF, the server extracts the text of every page and splits it into overlapping chunks of about one thousand characters. Each chunk is turned into a vector embedding with 768 dimensions and stored in PostgreSQL using the pgvector extension. Uploaded documents are deleted automatically after 24 hours.

How questions work

A question is embedded with the same model. The database returns the five closest chunks using cosine distance, which is the retrieval step of retrieval-augmented generation (RAG). The chunks are placed in a prompt and the language model, Gemini, writes an answer that is streamed back to the browser token by token using Server-Sent Events.

Limits and safety

Each visitor can keep three documents of up to 60 pages. Requests are rate limited and a global daily cap protects the free-tier AI quota. If the answer is not in the document, the assistant says so instead of guessing.